Topological Data Analysis for Fuel Consumption and Carbon Footprint Reducing in Large Vehicle Fleets

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Abstract. This paper presents a study on the optimization of fuel consumption and the reduction of Carbon Dioxide (CO_2) emissions for a large-scale logistics and transportation fleet. Facing the challenge of inaccurate direct emission measurements, this research employs Topological Data Analysis (TDA), specifically the Mapper algorithm, to uncover hidden patterns in high-dimensional operational data. The methodology involved the integration and cleaning of disparate datasets (*fuel orders* and *vehicle telemetry*), the application of a CO_2 estimation model, and the application of TDA to characterize vehicle performance and trip efficiency. Results reveal distinct topological structures correlating vehicle age and specific models with emission efficiency, identifying critical clusters of under performing units. The study demonstrates that TDA effectively segments complex logistical data, providing actionable insights for fleet renewal and operational adjustments to achieve net-zero carbon goals.

Keywords: topological data analysis · mapper algorithm · carbon footprint · vehicle fleet optimization

1 Introduction

Currently, air pollution and fossil fuel consumption represent both an economic and environmental concern for the transportation industry. With the Sustainable Development Goals (SDG) prioritizing global warming and the alarming increase of 80% in the production of CO_2 from 1999 to 2019 (Dai et al., 2023), corporations with extensive vehicle fleets have a major responsibility for contributing to the world's decarbonization. Nevertheless, achieving "net-zero" emissions requires more than simple fuel monitoring; the optimization demands a deep understanding of the non-linear relationships between vehicle characteristics, operational routes, and environmental impact.

Traditional statistical methods often struggle to capture complex global structures within noisy, high-dimensional data. Topological Data Analysis (TDA), a

technique rooted in algebraic topology, is particularly robust in analyzing systems; "data has shape, shape has meaning, and meaning brings value" (Carlsson, 2009). The use of TDA techniques has gained significant traction due to their ability to analyze non-linear systems through the generation of meaningful, simplified representations of such systems (Xu, 2024).

The application of topological analysis to the industry of transportation logistics, route optimization, and demand prediction has made valuable contributions to the literature. For instance, Esteve and Falco, 2025 applied TDA methods to classify trajectories to deepen the rationale behind complex spatial movement patterns. On the other hand, Rivera-Castro et al., 2019 designed a clustering method for time series based on TDA, from which a clusterwise regression method using matrix factorization techniques was employed; these methods allowed them to accurately predict the demand of different kinds of clients.

Route optimization for urban areas has also been explored in the literature. In Huang and Wu, 2024, the high dimensionality of Shenzhen citizens' spatiotemporal data was tackled by TDA techniques, from which behavioral transport patterns were found without reducing data complexity. While in Chen et al., 2019, citizens' spatiotemporal data from several cities is classified using a combination of TDA and time series analysis techniques.

Extending from the aforementioned applications, this paper aims to apply a combination of statistical and topological techniques to a fuel distribution partner company's fleet's operational data. By treating the fleet's high-dimensional logistical data as a topological space, we aim to identify significant and stable sub-groups of vehicles and trips that exhibit distinct emission behaviors using both Mapper algorithm for the most relevant variables and Takens embedding for the time series. These topological insights will be used to endorse actionable strategies and methodologies that will eventually reduce the logistics carbon footprint.

2 Preliminaries

Topological data analysis was introduced as such by Carlsson, 2009, proposing a new way of analyzing data that is commensurate with the complexity —high dimensionality, noise, and missing information— that data has acquired in the last decades, making use of geometry and topology. These methods and techniques deal with point clouds, i.e. finite sets of points equipped with a distance function (Carlsson, 2009), utilizing algebraic topology to identify persistent features —such as components, loops, and voids— that remain invariant under continuous deformations.

The following paragraphs describe the fundamentals of two classic TDA techniques: Taken's Embedding and Mapper Algorithm, used in this work.

Taken's Embedding. In practice it is rare to have an explicit mathematical description of a dynamical system of interest, instead, discrete observations can be made of some variables in different states of the system, resulting in a time

series (Perea, 2018). However, although a single time series may appear to be an oversimplification of the underlying dynamics, the Takens' embedding theorem (Takens, 1981) implies that they can actually be very useful.

Given time series data $f(t) = \phi(t)$ observed from a potentially unknown dynamical system (M, Φ) (where M is a smooth, compact, Riemannian manifold), the *sliding window embedding* of f, with parameters $d > 0$ (dimension) and $\tau > 0$ (delay), is the vector-valued function

$$\begin{aligned} SW_{d,\tau} f : \mathbb{R} &\longrightarrow \mathbb{R}^{d+1} \\ t &\longmapsto (f(t), f(t+\tau), f(t+2\tau), \ldots, f(t+d\tau)) \end{aligned} \tag{1}$$

where $d\tau$ is the window size. And for a time scope $T \in \mathbb{R}$, the *sliding window point cloud* is defined as

$$\mathbb{SW}_{d,\tau} f = \{SW_{d,\tau} f(t) : t \in T\} \tag{2}$$

Takens' theorem implies that the sliding window point cloud $\mathbb{SW}_{d,\tau} f$ provides a topological copy of $\Phi(t, p) : t \in T \subset M$ (the original system). In particular, this will reconstruct attractors (subsets of M that attract the evolution of states in close proximity), that can be used to identify trends in system behavior, such as periodicity. For further information, see Perea, 2018.

Mapper Algorithm The Mapper algorithm simplifies high-dimensional data into topological graphs by applying a *filter function* for dimensionality reduction (e.g. PCA, t-SNE, UMAP), partitioning the data into overlapping clusters (using HACA, DBSCAN, to name a few), and connecting —through edges— clusters —as nodes— that share common data points (Carlsson, 2020). This ease of visual interpretation makes the Mapper algorithm particularly valuable for data exploration and visualization.

Two ways of defining Mapper's algorithm are often distinguished: the topological version and the statistical version. In Algorithm 1 we present a brief overview of the latter, which is more straightforward for practical applications. A more detailed review of Mapper's topological definition, filter functions, and most commonly used clustering algorithms can be found as well in (Madukpe et al., 2025).

3 Methodology

3.1 Data Acquisition and Preprocessing

The study utilized two primary datasets spanning the years 2020 to 2024: *Fuel Orders*, which contains transactional data including requested liters, service dates, and locations; and *Vehicle Telemetry*, containing vehicle specifications, distance traveled, vehicles real fuel performance, and fuel transactions. Data preparation followed the CRISP-DM methodology. Key challenges addressed

Algorithm 1 Mapper Algorithm.

Require: A point cloud $X \subset \mathbb{R}^n$.
Require: Filter function (f), clustering algorithm ($\mathbf{C}$), percentage of overlap (ρ).
Apply $f : X \to \mathbb{R}^d$ to discriminate the characteristics of the dataset by projecting X to a lower space ($d < n$).
Build a cover on the image of $f(X)$, with a collection of open sets, $(U_i)_{i \in I}$ where I is a finite indexing set, overlapping according to ρ.
For each interval of cover U_i, cluster the points of the preimage $f^{-1}(U_i)$ using $\mathbf{C}$, to form a set of clusters C_{i_n}.
Each cluster in C_{i_v} forms a node in V.
Let $E = \emptyset$ be the set of edges.
for Each pair of clusters C_{i_v} and C_{j_u} **do**
    if $C_{i_v} \cap C_{j_u} \neq \emptyset$ **then**
        An edge connecting the two nodes is added to E.
    end if
end for
    return The topological graph $G = (V, E)$.

included high sparsity in "distance traveled" records and inconsistent vehicle identification.

To properly characterize the fleet, vehicles were identified by decoding their 17-character Vehicle Identification Numbers (VINs). Approximately 70% of the dataset had valid and correctly formatted VINs, which were used to extract the model year, manufacturer, and body type through the National Highway Traffic Safety Administration (NHTSA). For another 20% of the recorded data, manual extraction of the vehicle data could still be extracted, even with malformed identifier tags. Nevertheless, the remaining 10% were unrecoverable and thus unusable for the purposes of this study. This ultimately implies a successful recovery of metadata for 90% of the fleet. This is equivalent to 444 unique units, spanning 32 models (some of them described in Table 1) and 10 manufacturers.

In Table 2 the distances between most relevant zones are stated in kilometers. The zones represent different cities.

Table 1: Code names and descriptions of relevant models within the fleet.

Abbreviation	Model	Description
LDV-P	Light Duty Vehicle: Pickup	Light utility vehicle with open cargo bed, 4x4 drive, and capacity less than 1.5t.
HDV-T	Heavy Duty Vehicle: Tanker	Heavy-duty truck, rigid chassis class 7/8, optimized for fluid transport.
LCV-V	Light Commercial Vehicle: Van	Light commercial van type, urban design, reduced volumetric capacity.
LCV-L	Light Commercial Vehicle: Large Van	Large/full-size light commercial vehicle.

Table 2: Distance from zones in kilometers.

	Zona 1	Zona 2	Zona 3
Zona 1	0	502	710
Zona 2	502	0	209
Zona 3	710	209	0

3.2 Feature Engineering: CO_2 Estimation

Because the generation of CO_2 depends on the type of fuel used, vehicles were classified into two categories: gasoline vehicles and diesel vehicles. Additionally, with the purpose of obtaining a target variable that more closely reflects reality, real-world fuel efficiency and trip distance are used to approximate the produced CO_2 emissions.

Equation (3) was applied to gasoline vehicles. According to EIA (2024), 8.49 is the number of kilograms of CO_2 consumed per gallon of gasoline. Since units are different, a conversion was needed with a constant of $3.78541 \frac{L}{gal}$.

$$CO_2 = \frac{\text{Distance} * \frac{8.887}{3.78541}}{\text{Fuel Performance}} \tag{3}$$

On the other hand, Equation (4) was applied to diesel vehicles. According to EIA (2024), 10.19 is the number kilograms of CO_2 consumed per gallon of diesel. Since units are different, a conversion was needed, using the same constant as above.

$$CO_2 = \frac{\text{Distancia} * \frac{10.180}{3.78541}}{\text{Fuel Performance}} \tag{4}$$

3.3 Topological Analysis

To analyze the high-dimensional and non-linear relationships within the fleet's operational data, we employed two primary mathematical frameworks, as mentioned above. Takens Embedding (see Section 2) was used to reconstruct the phase space of the dynamical system from a sequence of observations, as the study found pertinent to either confirm or reject the existence of an underlying periodicity or cyclic stability in distance and emission time-series, since it could reveal periodic peaks in CO_2 emissions. To find the optimal τ (deelay) the first local zero-crossing of the Autocorrelation Function (ACF) (see Fraser and Swinney, 1986) was used. Also, to find the optimal d (embbeding dimension) the False Nearest Neighbour method (see Wallot and Mønster, 2018) was used.

The Mapper Algorithm (see Section 2) was used to build a simplicial complex or graph representation, capturing the global shape of the dataset. The hypothesis is that by exploring the underlying topology of the vehicle fleet data, separability can be found, and in turn, the specific study of each cluster can provide information on the behavior of the vehicles.

3.4 Experimental Design

Three experimental scenarios were designed to analyze different logistical aspects of the fleet, such as temporal stability (described in Table 3), trip segmentation (described in Table 4), and fleet characterization (described in Table 5). The optimal parameters of each technique were obtained via grid search through multiple experiment iterations to achieve useful visualizations. The results of these experiments are explained in section 4.

Table 3: Experiment A. Temporal Stability (Takens Embedding).

Dataset	Vehicle Telemetry and Fuel Orders (Time-series aggregated).
Variables	Estimated CO_2, and Distance.
Description	Before searching for structural clusters, a Takens embedding was applied to evaluate the periodicity of the system and determine if the fleet's emissions and travel distances exhibit predictable, cyclic behaviours over time.

Table 4: Experiment B. Trip Segmentation (Logistics View).

Dataset	Vehicle Telemetry (Cleaned).
Variables	Liters supplied, distance, estimated CO_2, truck type, zone and destiny.
Parameters	UMAP (2D) with identity as the filter function.
Description	Knowing beforehand that the CO_2 variable depends on not only the distance and and liters supplied, but also in the kind of vehicle and kind of road. Two instances were designed with the same filter and parameters, but with a different selection of variables. The first instance, using only CO_2, distance and liters supplied, aims to identify general "trip types" without considering their start and ending points. In the second instance, all variables described above were used, allowing us to characterize each trip according to its destination and, more specifically, the type of vehicle utilized. For categorical variables, ordinal encoding was applied based on the database sorted alphabetically by zone and then by destination; this was done to loosely preserve geographical information, while ensuring nearby destinations received similar values after encoding.

4 Results

4.1 Experiment A

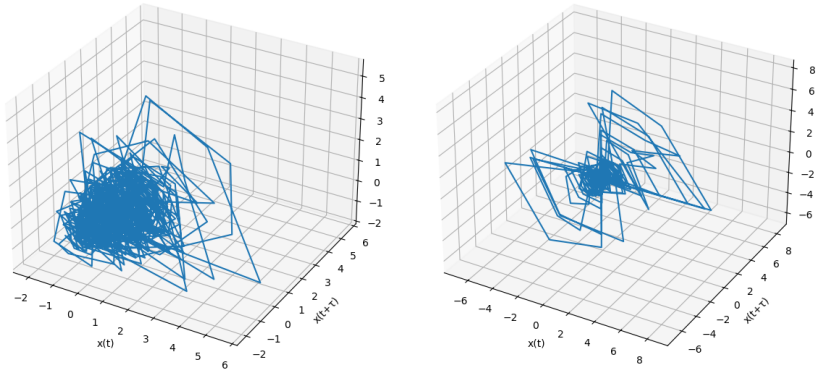
Regarding CO_2 variable, the representation in the reconstructed space did not show a structured pattern (as seen in Figure 1-a). The data points were distributed in a chaotic manner, failing to form cycles or defined structures (i.e.

Table 5: Experiment C. Fleet Characterization (Asset View).

Dataset	Vehicle Telemetry and Fuel Orders (Aggregated by VIN).
Variables	Liters supplied, Distance, Zone, Destination, Vehicle Class (pickup, van, truck) and Estimated CO_2
Parameters	UMAP (2D) with CO_2 projection as the filter function.
Description	In this second set of variables, additional parameters were added to characterize each trip according to its destination and the type of vehicle used explicitly. An ordinal encoding was applied to the new categorical variables. For the distance and zone variables, the encoding priority was based on the database sorted alphabetically by zone and then by destination; this was intended to vaguely preserve geographic information, ensuring that nearby destinations would have similar values after encoding.

distinguishable attractors). This was mainly attributed to data quality, as subsequent analyzes revealed that certain values were inconsistent or demonstrated significant irregularities. Consequently, the results were unreliable, preventing the obtainment of useful insights regarding the temporal dynamics of CO_2 consumption.

Regarding the distance variable, the embedding displayed a more compact point cloud with slightly more organized trajectories (see Figure 1-b). However, the observed periodicity was minimal and insufficiently consistent to confirm the existence of a genuine cyclic pattern. In other words, although some regularity appeared to exist, it was neither clear nor constant across different months of the year; therefore, it cannot be concluded that the system performed periodic dynamics.

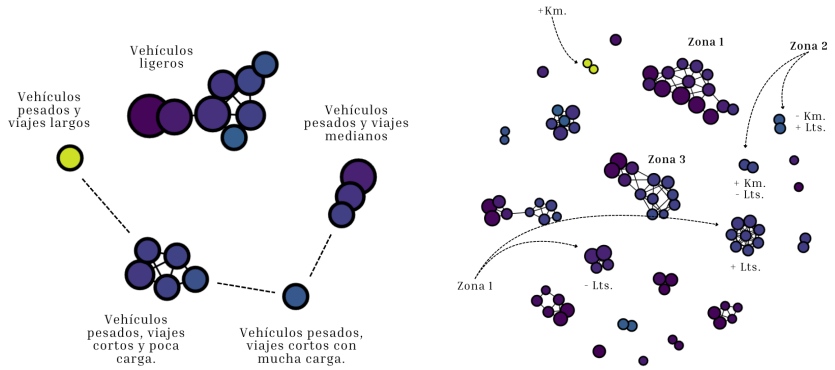


(a) For CO_2 ($\tau = 1, d = 10$) (b) For distance ($\tau = 1, d = 10$)

Fig. 1: Takens embedding for CO_2 and distance traveled time series.

4.2 Experiment B

Figure 2 shows the graphs resulting from each instance, along with a brief description of each component. Broadly speaking, both components of each instance are primarily determined by trip destinations. This is expected, as similar destinations imply comparable distances and CO_2 emissions. In certain instances, however, there is still variability in the quantity of liters delivered within the same destination.



(a) Instance 1: Three variables. (b) Instance 2: Six variables.

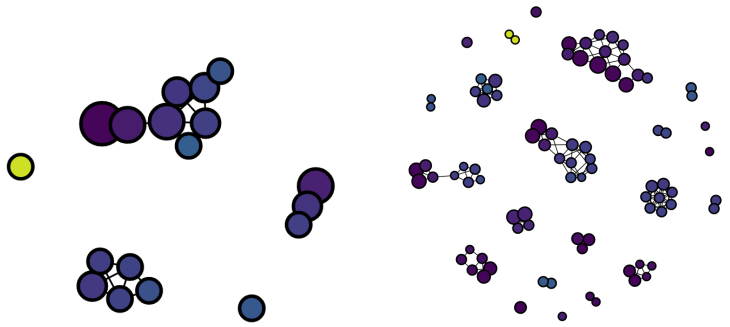
Fig. 2: Graphs generated through Mapper algorithm using UMAP 2D as filter and CO_2 as the color scale to describe the types of trips. Labeled manually according to the characteristics of each component.

As can be visualized in Figure 2.a, the first instance showed 5 connected components, Component A corresponds to trips involving light vehicles. That is, those where less than 1,000 liters were transported, according to our feature engineering criteria. On the other hand, the remaining connected components consist of trips made with heavy vehicles. Unlike the light vehicle trips, these exhibit enough variability to be split into several distinct connected components. Among these, Component B stands out; since it represents trips with CO_2 emissions significantly higher than the rest.

The second instance (Figure 2.b) displays connected components primarily determined by trip destinations. This is expected, as similar destinations imply similar distances and CO_2 emission levels. Note that in some cases, there exists variability in the volume of liters delivered for the same destination.

It is important to note that in this graph, despite consisting of several more connected components, a specific component containing the highest emissions persists.

Given that routes (trips sharing a destination) represent a significant grouping from a topological perspective, it is valuable to conduct a detailed study of the internal behavior of each route. This will help determine which ones are most problematic in terms of either CO_2 emissions or logistics.



(a) Graph with three variables. (b) Graph with six variables.

Fig. 3: Mapper graphs about the dataset *Vehicles Telemetry*

4.3 Experiment C

Applying the Mapper algorithm as described in Table 5, six connected components were obtained representing the topology of the vehicle data in the fleet, originally in six dimensions. From left to right and from top to bottom, as shown in Figure 5, they are described below. Also, a labeled graph is shown in Figure 4.

Component A. It includes exclusively —and almost all— trucks and tractors, of different models. Through coloring by different variables (see Figure 5) a certain polarization can be observed within the component: When coloring by CO_2, there is a decreasing trend from left to right, in contrast to the inverse trend that occurs when coloring by year of manufacture of the vehicle. The same happens with fuel efficiency. That is, newer trucks and tractors have better fuel efficiency, and consequently also lower CO2 emissions.

Component B. This component contains exclusively all HDV-T models from 1984 and 1981; the oldest in the fleet. Despite being the same model and year, analyzing the content of the component's single node reveals notable variations in vehicle performance; CO_2 emissions range from 0.22 to 2.44 kg/km. It can

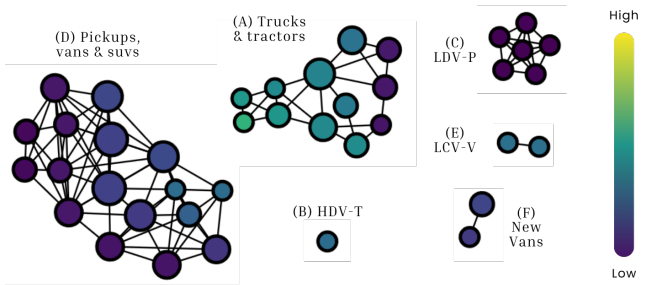


Fig. 4: Graph generated through Mapper algorithm to describe the fleet's structure, labeled manually according to the data in each cluster. Color corresponds to CO_2/Km.

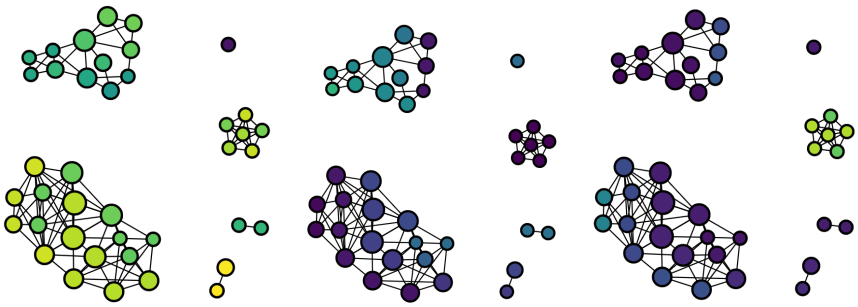
be inferred that older models exhibit greater variability in performance and emissions due to their years of operation and varying levels of maintenance.

Component C. It includes exclusively LDV-P. The shape of the connected component, with six fully interconnected and nearly homogeneous nodes, reflects the reliability of this vehicles (at least in this study). Even though this component includes models manufactured between 2013 and 2022, their CO_2 emissions remain consistently low.

Component D. This is the largest component, as it includes pickups, vans, and SUVs. The same trends observed in Component A are present here: Nodes with lower average CO2 emissions tend to have a more recent average manufacturing year. Within this component, there is another large cluster of LDV-P, but unlike those in Component C, their CO2 emissions are slightly higher, even though they are distributed across the same manufacturing years. The cause of these differences between the two groups is likely due to the improved maintenance provided to some vehicles.

Component E. This component includes exclusively LCV-V models from 2009 and 2007, which have lower performance and higher emissions than other similar vehicles such as LCV-L, which are more convenient in this respect.

Component F. This component includes various models from recent years, including a 2021 LCV-V, which show acceptable performance and emissions, by van standards.



(a) Color: Vehicle age. (b) Color: CO2/Km. (c) Color: Vehicle efficiency.

Fig. 5: Graphs generated through Mapper algorithm to describe the fleet's structure, using color scaling to remark relevant variables.

5 Conclusion

None of the two tests made with Taken's Embedding provided strong evidence of periodicity. For CO_2, this was due to data quality issues; while for traveled distance, although a slight structure was present, it was insufficient to prove a cyclic behavior.

On the other hand, the application of Mapper algorithm allowed for the detection of non-trivial patterns that aggregate statistics missed. It was successfully identified that longer trip distances involves higher CO_2 emissions; newer vehicles show greater fuel efficiency and a reduction in CO_2 emissions per kilometer traveled, as observed in the trucks and tractor-trailers of components A and B of the Mapper analysis; and older vehicles exhibit greater dispersion in their emissions and efficiency characteristics, suggesting that their performance depends heavily on individual maintenance

The results, combined with fleet operational information and exploratory data analysis, allowed for more informed decision-making regarding vehicle renewal, maintenance, and operation; aligning sustainability goals with technical and financial criteria. Topological Data Analysis has demonstrated to be a novel and useful tool for data analysis, allowing for the identification of patterns in databases with several variables; however, such utility is reduced if it is not combined with proper data processing and statistical analysis that complements the understanding of the data.

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